AI AND ML DRIVEN MAINTENANCE SERVICE PROVIDER MATCHING PLATFORM FOR SRI LANKA

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

This report presents the integrated research prototype for an AI and ML driven maintenance service provider matching platform for Sri Lanka. The common problem is selecting relevant and operationally suitable providers from heterogeneous profile, document, schedule and review information. Four components address this problem through hybrid recommendation, demand prediction and provider filtering, document-based provider scoring, and review sentiment and credibility analysis. The intended customer workflow connects up to twenty initial candidates to up to ten schedule-eligible candidates and a final shortlist of up to five, with onboarding evidence and weekly-demand views as supporting branches. The recommendation component combines lexical, multilingual semantic and interaction-based scores and reports Precision at twenty of 0.9442 and NDCG at twenty of 0.9458. The demand component reports XGBoost R squared of 0.882890 on synthetic order counts, while document scoring reports R squared of 0.994 on generated policy labels. The review module reports mean aspect sentiment accuracy of 88.94 percent and review-credibility accuracy of 96 percent. These values describe separate experiments and are not combined into an overall platform accuracy. The common contribution is a modular selection framework with shared identities, explicit interfaces and distinct evidence meanings. Selected database demonstrations support component integration, but a versioned same-request trace and measured final-ranking benefit are not supplied. Synthetic targets, category differences, permissive fallbacks and missing stakeholder validation constrain operational conclusions. Future work prioritises consistent contracts, strict input and state handling, independent data evaluation and a controlled user pilot.

Keywords: service provider matching, hybrid recommendation, provider evidence, review intelligence, component integration

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We acknowledge the researchers and developers of the algorithms, software libraries and technical documentation cited in this report. AI assistance in preparing the common document is disclosed in Appendix C. The team retains responsibility for confirming the final technical account, individual and shared contributions, and submission details.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
Bi-LSTM	Bidirectional long short-term memory
CF	Collaborative filtering
CSV	Comma separated values
CV	Cross-validation
MAE	Mean absolute error
MAPE	Mean absolute percentage error
ML	Machine learning
NDCG	Normalized discounted cumulative gain
OCR	Optical character recognition
PDF	Portable Document Format
RF	Random Forest
RMSE	Root mean squared error
RTDB	Realtime Database
SDG	Sustainable Development Goal
TF-IDF	Term frequency–inverse document frequency
UI	User interface
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

The Maintenance Service Provider Matching Platform for Sri Lanka applies artificial intelligence (AI) and machine learning (ML) to select suitable professionals from a large collection of service listings. Customers seeking an electrician, plumber, cleaner, carpenter or maintenance technician need to understand both the relevance of a provider's services and the evidence supporting that provider's suitability. A category label and an overall rating provide useful starting points, but they leave important questions unanswered. The customer may describe a problem using different words from the provider's profile, require work during a particular interval, or need to distinguish comments about workmanship from comments about punctuality.

The four individual theses define complementary responses to these questions. A hybrid recommendation engine interprets a natural-language requirement and ranks providers. A demand and filtering component supplies category-level planning information and checks declared schedules and geographical proximity. A document-based credibility component converts onboarding evidence into structured provider information. A review-intelligence component analyses sentiment and review credibility before refining a candidate list [1]–[4]. Their common purpose is to make provider selection more informed while reducing the amount of disconnected information that customers must interpret.

The project is situated in the Sri Lankan maintenance-service context described by the group. The intended users include customers with household or technical maintenance needs, providers who maintain professional profiles, and administrators who oversee evidence and application records. The supplied documents establish this application setting and report technical experiments. They do not contain a nationally representative survey of provider discovery or a measured estimate of the economic cost of current practice. The report therefore motivates the project through its identified decision problems without assigning unsupported market statistics.

Machine learning contributes to different parts of the solution in different ways. Text similarity supports initial retrieval, regressors approximate generated numerical targets, and a neural model classifies review sentiment. Schedule and distance filtering remain deterministic operations. This combination allows the platform to use a method appropriate to each task rather than assume that every decision requires a trained model. The main research question at group level is whether these distinct outputs can form a coherent and inspectable provider-selection workflow.

1.2 Stakeholders and user needs

Customers need a shortlist that relates to the actual service requirement, a clear indication of declared schedule eligibility, and understandable evidence about the providers shown. They also need to recognise when a result contains incomplete information. A customer should be able to distinguish a highly relevant profile from a confirmed appointment, and a document-derived score from an independently authenticated qualification. These are design requirements arising from the documented workflow; a completed stakeholder study has not established their relative importance [1]–[4].

Providers need a way to represent their service descriptions, working hours, locations, qualifications and experience. The document component supports structured onboarding evidence, while the weekly-demand interface offers a view of category-level activity under a defined prediction scenario. Providers also need opportunities to correct information. An incorrect category mapping, an unreadable certificate or a misunderstood review can affect their presentation even when the software completes processing successfully.

Administrators need to inspect processing outcomes, identify failures and maintain consistent records across the four components. The application materials described in the demand thesis include customer, provider and administrator interfaces, a demand timetable and filtering-history views [2]. Administrative responsibilities extend beyond viewing a score: they include handling uncertain evidence, controlling access to personal documents, resolving category discrepancies and preventing stale results from appearing current.

The research team and supervisors need traceable evidence for the claims made about the integrated project. Component metrics, database demonstrations and proposed tests have different evidential roles. This report preserves those roles and provides a contribution matrix so that individual responsibilities remain visible within the common solution. It does not treat access to a shared frontend or database as proof that one member authored all associated work.

1.3 Alignment with the Sustainable Development Goals

The project's principal intended alignment is with Sustainable Development Goal (SDG) 8, which addresses decent work and economic growth. Target 8.3 concerns productive activities, entrepreneurship and the development of smaller enterprises [5]. A platform that helps skilled providers present services and evidence could contribute to that pathway. The relevant mechanism is improved access to service opportunities through clearer matching and representation, rather than an assumption that a prediction score itself creates employment.

Sustainable Development Goal 9 provides a secondary alignment through its attention to innovation and technological capability [6]. The project develops a modular digital service workflow using established analytical methods and shared application infrastructure. This is a research and product-development contribution. It does not demonstrate an improvement in national infrastructure or a measured expansion of digital access.

Both alignments require inclusive implementation. Documentation-based scores may disadvantage competent providers with limited formal certification. Historical interaction signals may favour established providers, and review models may perform differently across writing styles or languages. The current experiments do not measure income, employment quality or exclusion. Section 7.5 therefore proposes indicators that could evaluate benefits and adverse effects during a controlled pilot.

1.4 Purpose and scope of the common report

This report presents the common problem, the division into four subproblems, the solution architecture, shared interfaces, implementation evidence and overall

evaluation. It follows the eight-chapter structure supplied in Common Report Content.docx [7]. The individual theses remain the detailed accounts of each member's datasets, algorithms, model configurations and component experiments [1]–[4]. Here, those details are selected and synthesised where they explain the complete system or its integration requirements.

The report draws on all four supplied theses. Component results are attributed to their original reports; no new model training, live Firebase transaction, user recruitment or complete application deployment was performed to prepare this common report. Arithmetic comparisons are identified where they help interpret reported values. Architecture diagrams are redrawn as a common-system synthesis, and intended cross-component handoffs are distinguished from recorded local processing paths.

The scope includes the customer-selection workflow and its supporting onboarding and demand branches. Payments, dispute resolution, authority-level identity authentication and a booking-conflict calendar are outside the demonstrated scope. The common contribution is assessed as a research prototype with documented component capability and an identifiable integration design. Missing evidence is discussed as a validation limitation rather than replaced with an invented group result.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem is that customers cannot consistently translate a maintenance requirement into a shortlist of relevant, operationally suitable and well-supported service providers using disconnected profile, document, schedule and review information. Existing information may be available, but its meaning and relationship to the current request are difficult to establish. A provider can match a keyword while offering the wrong specialisation, have an appropriate profile while being outside the requested working interval, or have a high overall rating without strong feedback on the aspect the customer values.

The problem also affects providers and administrators. Providers must communicate evidence that varies in format and completeness, while administrators need to convert that evidence into consistent records. Category-level planning information introduces another question: what activity is expected for a service over the coming week? This is related to marketplace operation, but it has a different prediction target from the suitability of an individual provider for one request.

The research consequently treats provider selection as a sequence of decisions supported by several kinds of evidence. The platform needs to connect semantic relevance, declared operating context, onboarding evidence and previous customer feedback without confusing their meanings. A combined interface is useful only if the identities, service categories and processing states remain consistent as information moves between components [1]–[4].

2.2 Root causes and overall research gap

The first cause is variation in how customers and providers describe the same service. Direct word overlap preserves explicit terms but may miss a relevant description that uses different wording. Sentence representations offer a complementary way to compare meaning. Sentence-BERT research provides the methodological basis for comparing sentence embeddings using cosine similarity [8].

The recommendation thesis combines lexical and semantic relevance with a separate history and provider-quality signal [1].

The second cause is the separation between profile relevance and operational suitability. A relevant provider may have a schedule that does not cover the requested interval or a location that makes the service inconvenient. The filtering component addresses declared working hours and great-circle distance, but the individual thesis identifies permissive handling of malformed times and missing coordinates [2]. Reliable integration therefore depends on input validation as well as on the presence of a filter function.

The third cause is heterogeneous evidence. Identity files, registration documents, certificates, experience letters and portfolios describe different claims. Text extraction can transform them into inputs, but recognising text does not establish document authenticity. The provider-credibility thesis identifies the need to distinguish uploaded evidence, successful extraction, policy scoring and verification [3]. Combining these states into one badge can conceal uncertainty from downstream consumers.

The fourth cause is the limited meaning of aggregate review statistics. Reviews can contain different opinions about Quality, Punctuality, Communication and Professionalism, while unusual review behaviour may affect how strongly a review should influence evaluation. The review module treats sentiment and anomaly-based credibility as separate analytical processes [4]. Statistical anomaly detection, however, cannot establish fraudulent intent on its own [9].

The overall gap is an application-specific integration and validation gap. The four methods exist as component contributions, but the complete system needs consistent identifiers, category translations, candidate handoffs and evidence semantics. The supplied reports do not provide one versioned trace showing all four components operating on the same request and provider records. The group contribution is therefore assessed through the coherent architecture and reported component evidence, with complete workflow validation remaining a defined next step.

2.3 Subproblem and component mapping

Table 2.1 defines the four component responsibilities used throughout this report. Component numbering follows the customer workflow: initial recommendation, contextual filtering and planning, supporting provider evidence, and review-based refinement. Document assessment normally precedes customer requests as part of onboarding. Demand forecasting operates alongside request processing and supplies planning information rather than the current filter's ranking score.

Table 2.1 Four subproblems and component responsibilities

Component	Distinct subproblem	Main output
C1 Hybrid recommendation	Relate a natural-language requirement to relevant provider profiles using lexical semantic and history signals.	Ranked Top-20 candidates and recommendation scores.
C2 Demand and filtering	Represent category-level activity and refine supplied candidates using declared schedules distance and weather context.	Seven-day category forecasts and up to ten eligible candidates.
C3 Provider credibility	Convert heterogeneous onboarding documents into consistent evidence features and a defined policy score.	Provider feature payload score and level.
C4 Review intelligence	Interpret service-aspect sentiment and review credibility for an existing candidate set.	Provider review evidence and final Top-5 reranking.

The boundaries prevent duplication of the term credibility. C1 uses rating, booking success and interaction frequency as part of its own scoring mechanism. C3 produces a document-derived policy score. C4 estimates the reliability of review evidence through anomaly analysis. These three meanings refer to different inputs and outputs. The common architecture preserves them separately; it does not assume that C1's collaborative score already consumes C3's output [1], [3], [4].

2.4 Overall aim and group objectives

The overall aim is to develop and evaluate an integrated maintenance-service selection prototype that connects request relevance, operational context, provider evidence and review intelligence within one service platform. The intended outcome

is an inspectable shortlist that supports a customer's choice and provides associated planning and evidence information to providers and administrators.

Six group objectives translate that aim into assessable responsibilities. Table 2.2 identifies the output and evidence for each objective. These are the objectives synthesised for the common report. Their assessment uses the supplied research evidence, while the stronger release criteria in Chapter 6 are proposed criteria for a subsequent integrated evaluation.

Table 2.2 Common objectives and evaluation evidence

Objective	Required outcome	Evidence used
G1 Request relevance	Generate ranked candidates from the service requirement.	C1 method comparison using retrieval and ranking metrics.
G2 Context and planning	Produce forecast summaries and evaluate schedule and distance conditions.	C2 regression results forecasts and helper checks.
G3 Provider evidence	Convert documents into a bounded score with structured inputs.	C3 model results document fixtures and database example.
G4 Review refinement	Classify aspect sentiment assess review anomalies and rerank candidates.	C4 aspect results credibility report and workflow description.
G5 Integration consistency	Preserve identities categories candidate lineage and output semantics across components.	Contract synthesis and documented integration boundaries.
G6 Responsible evaluation	Distinguish measured capability from unsupported outcome and deployment claims.	Common evidence review limitations and validation framework.

The objectives deliberately include consistency and responsible interpretation. Strong model performance would not satisfy G5 if a category translation silently changes the intended service or if a request loses its candidate association. G6 is also necessary because the intended users must understand the distinction between predicted demand, declared eligibility, document evidence and review-derived estimates.

2.5 Scope assumptions and constraints

The platform is designed around maintenance and related service categories, registered users and provider profiles. The demand and review components each represent fourteen service categories, while the document-scoring experiment uses ten categories [2]–[4]. The report does not claim identical coverage across all models. A common category catalogue must preserve original service labels and map them to each component's supported vocabulary. Unsupported categories require an explicit response rather than a silent assignment to another service.

The component experiments use different data and evaluation units. C1 reports provider, request and interaction datasets and evaluates category-level relevance. C2 uses 3,000 synthetic demand records. C3 uses 2,000 synthetic provider records. C4 reports 25,000 annotated reviews [1]–[4]. These records cannot be combined into one sample size for an integrated study, and their metrics cannot be averaged into a meaningful overall system accuracy.

The common application data layer is described as Firebase in the implementation sections of all four theses. Some C1 methodology passages instead refer to MongoDB [1]. This inconsistency is recorded in Appendix C. The architecture uses Firebase as the shared application interface because it is the repeatedly described implementation boundary, while a confirmed deployment manifest is still needed to resolve any experimental or alternate database role.

The current filter checks declared weekly working hours rather than a confirmed booking calendar. It uses spherical distance rather than road travel time, and weather is an advisory. Document recognition does not authenticate the issuing authority. Review anomalies do not prove that a review is fake. Forecast counts are scenario outputs built from fixed future inputs rather than promises of orders. These assumptions define the present prototype and the limits of its user-facing interpretation.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The integrated solution has a customer-selection path and two supporting operational branches. In the selection path, the customer submits a natural-language maintenance requirement. C1 produces an initial ranked provider set, C2 evaluates the candidates against declared schedule and location information, and C4 uses customer-review evidence to refine the resulting set. C3 supplies structured onboarding evidence through provider records. C2's forecasting branch supplies category-level information to provider and administrator views [1]–[4].

This arrangement connects the components through shared records rather than forcing all analytical logic into one model. A provider identifier links profile information, uploaded evidence, candidate entries and reviews. A request identifier distinguishes one customer's service need from another. These associations enable each component to retain its own internal method while contributing an output that the platform can interpret.

The intended customer outcome is a final shortlist of up to five providers with understandable supporting information. The reported component cardinalities support an intended Top-20 to up-to-10 to up-to-5 sequence. That sequence reconciles C1's Top-20 output, C2's limit of ten eligible candidates and C4's Top-10 input and Top-5 output. It is an integration design derived from the supplied documents, not a newly observed full-system test. Chapter 4 explains what must be preserved to make the handoffs reliable.

The planning branch should remain visible as a separate function. Weekly category forecasts may help providers and administrators inspect potential activity. The current filtering implementation does not use those forecasts to order candidates, and its weather request is not a live covariate feed into the demand regressor [2]. Keeping that relationship explicit prevents the common report from attributing an unimplemented optimisation mechanism to the platform.

3.2 High level architecture

Figure 3.1 presents the shared application boundary and the four component services. The client layer supports customer requests, provider profiles and administrative inspection. Firebase supplies the shared provider, request, review and demand records described in the theses. Python-based analytical services perform the modelling and worker operations, while document storage and weather acquisition provide supporting inputs [1]–[4].

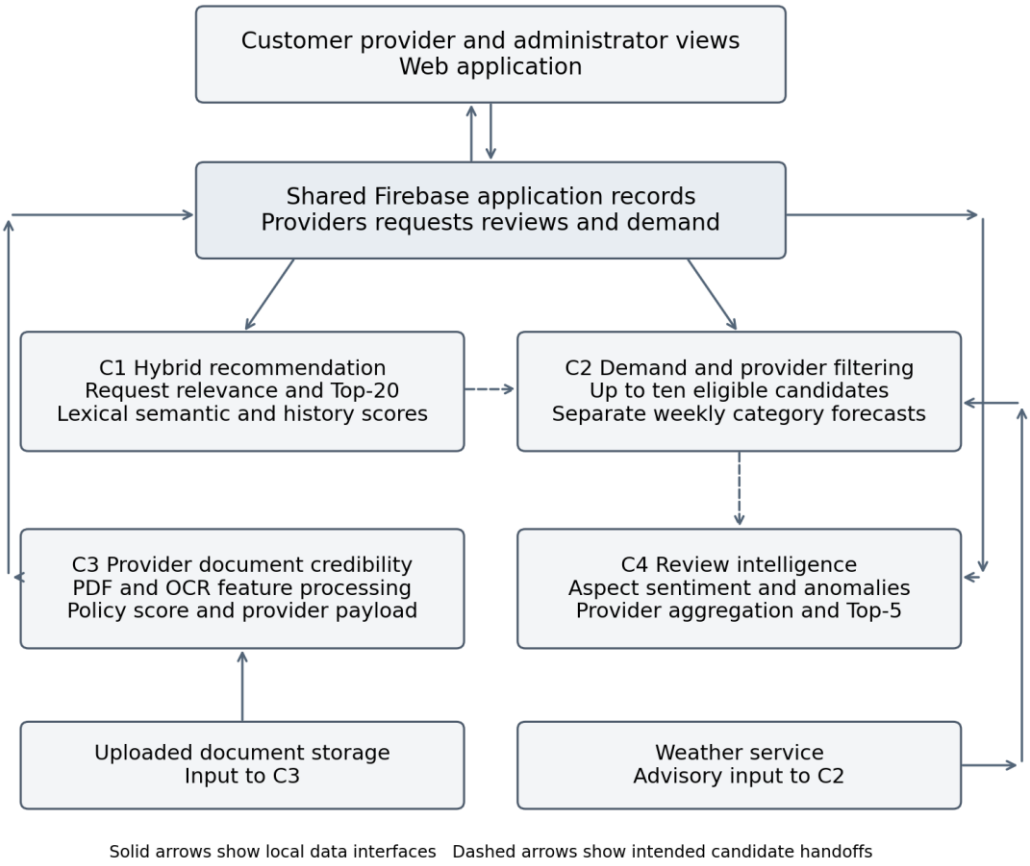


Figure 3.1 Common application and component architecture

The diagram distinguishes documented local interfaces from the intended orchestration between C1, C2 and C4. The theses describe component-level database interactions and individual processing paths. They do not identify one deployed orchestration service, exact endpoint catalogue or release version containing all handoffs. The common orchestration responsibility represented in this architecture concerns the function required to associate a request, its candidate sets and final

response; it is not a claim that a particular coordinator implementation was independently verified.

C3 operates asynchronously from customer ranking. Its result belongs to the provider record and can support administrative inspection or an evidence summary. The common design does not introduce a new undocumented numerical dependency from C3 to C1. If the team later decides to use document scores as a recommendation feature, that change needs an explicit interface, a calibrated policy and a separate evaluation of how it affects relevance and inclusion.

3.3 Component 1 summary

C1 is the Hybrid Multialgorithm Recommendation Engine, attributed in the supplied thesis to Hewapathirana N.T., student IT22371836. Its subproblem is matching a customer's natural-language request to provider descriptions while incorporating available provider-quality and interaction information. The inputs include provider descriptions, service categories, ratings, booking outcomes, interaction counts and user history [1].

The method combines term frequency–inverse document frequency (TF-IDF) cosine similarity with embeddings from paraphrase-multilingual-MiniLM-L12-v2 and a custom credibility-aware collaborative mechanism. The official model card describes a multilingual sentence-transformer model; this report uses the term sentence embeddings to avoid treating the component as a newly trained full BERT model [10]. The internal provider-quality calculation uses weights of 0.50 for rating, 0.30 for booking success and 0.20 for interaction frequency, together with a history adjustment where available.

After normalization, the hybrid fusion assigns 0.30 to TF-IDF, 0.35 to semantic similarity and 0.35 to the collaborative score. The component returns Top-20 candidates. Its reported evaluation samples 500 requests from the request dataset and defines relevance by equality between request and provider service categories. The hybrid obtains Precision@20 of 0.9442 and normalized discounted cumulative gain at twenty (NDCG@20) of 0.9458 [1]. These are category-ranking results; they do not measure customer satisfaction or maintenance outcomes.

The output contributes the initial relevant candidate set. Its usefulness to the complete platform depends on preserving provider identifiers and recommendation scores when the filtering stage changes the order. The lexical, semantic and history scores remain distinct from C3's document policy score and C4's review credibility output.

3.4 Component 2 summary

C2 is the Service Demand Prediction and Provider Filtering System. The supplied thesis leaves the member's name and student ID blank, so authorship is identified here by the component responsibility rather than inferred from computer paths or account names [2]. It addresses two related operational questions: category-level activity over a coming week and declared suitability of candidates for a requested interval and location.

The demand branch compares Random Forest and XGBoost on 3,000 generated records spanning fourteen categories and five districts. Date engineering and target encoding produce 25 model inputs. The seed-42 random split contains 2,400 training and 600 holdout rows. XGBoost obtains R squared 0.882890, root mean squared error (RMSE) 3.423339 and mean absolute error (MAE) 2.926993 in generated order-count units. Derived demand-band accuracy is 79.33% [2]. Random Forest and XGBoost are established ensemble methods [11], [12]; their project-specific performance remains conditional on the synthetic target and validation design.

The forecasting function produces seven dates for fourteen categories using saved model, encoder and column-order artefacts. A comma separated values (CSV) file pair for 3–9 August 2026 contains 98 daily entries and totals 1,926 predicted orders. These are scenario results from fixed inputs, not observed future bookings. The shared forecast payload supports a weekly timetable and provider-category demand view [2].

The filtering branch reads request and provider records, computes schedule eligibility, orders eligible providers by Haversine distance and retains up to ten. Open-Meteo supplies weather data, while the project's own rainfall rules produce the advisory [13]. Weather advice does not currently change candidate inclusion or

order. The component connects initial relevance to operational context, but input validation, date coverage and preserved candidate lineage are necessary for reliable combined operation.

3.5 Component 3 summary

C3 is the Provider Credibility Prediction System, attributed to EMNS Roopasinghe, student IT22117182. It converts identity, registration, qualification, experience and portfolio evidence into a provider assessment. Native Portable Document Format (PDF) extraction precedes optional optical character recognition (OCR), after which keyword and regular-expression rules derive model inputs [3].

The modelling experiment uses 2,000 synthetic provider records and ten predictors. A capped weighted policy generates the target score. Random Forest and XGBoost are tuned through three-fold cross-validation and evaluated on a 400-record seed-42 holdout. XGBoost records RMSE 1.141 score points and R squared 0.994, compared with 1.569 and 0.989 for Random Forest [3]. The deterministic target formula reproduces the generated labels exactly, so the regression result represents approximation of that policy rather than independent validation of provider reliability.

The score is bounded from zero to one hundred. Beginner applies below 60, Intermediate from 60 to below 85, and Professional at 85 or above. These are application presentation bands, not official qualification classes. Four document fixture demonstrations produce structured outputs, and one recorded integration example processes five documents and writes a score of 84.60 with an Intermediate level to Firebase [3].

The component contributes onboarding evidence and an inspectable assessment payload. Its current processing and verification flags require refinement because file presence and completed inference can be treated as verification. The common design preserves document-derived evidence as a supporting branch and requires uncertain extraction to be visible before any stronger verification claim reaches the customer interface.

3.6 Component 4 summary

C4 is the Bi-LSTM Based Sentiment Analysis and Review Credibility Module, attributed to Tharushika W.A.M., student IT22914200. It receives a candidate provider set and analyses associated reviews. Its purpose is to represent separate service-quality dimensions and the estimated credibility of the review evidence before final provider refinement [4].

The sentiment method is a multi-task bidirectional long short-term memory network (Bi-LSTM) with a custom attention layer. A vocabulary limit of 15,000 and sequences of 80 tokens feed a 128-dimensional embedding, a bidirectional layer with 96 units, a shared dense layer with 96 units, dropout and four three-class output heads. The outputs represent Positive, Neutral and Negative sentiment for Quality, Punctuality, Communication and Professionalism. The shared neural representation allows the four tasks to use common review information while retaining separate predictions [4].

The thesis reports 25,000 annotated reviews across fourteen categories, partitioned into 17,466 training, 3,761 validation and 3,773 test instances. The four aspect accuracies are 96.10%, 87.73%, 86.35% and 85.58%, with an arithmetic mean of 88.94%. A separate Isolation Forest process supplies an anomaly-related review credibility signal. Its reported overall accuracy is 96%, with weaker F1 for the minority class labelled Fake than for the class labelled Real [4].

Review-level sentiment and credibility outputs are associated and aggregated by provider. The module is designed to rerank an initial set of ten candidates and return five. The thesis does not publish the exact provider-level aggregation formula or final reranking coefficients, so this report describes the process without inventing weights. The component contributes detailed customer-feedback evidence; anomaly output remains a statistical signal rather than proof of deceptive intent.

3.7 End to end user and data flow

Figure 3.2 shows the intended candidate handoff together with the supporting branches. Before a customer request, a provider creates a profile and may submit documents for C3 processing. The resulting evidence payload is stored against that

provider. Customer reviews and interactions are also associated with provider identities. This accumulated information establishes the shared context for later selection [1], [3], [4].

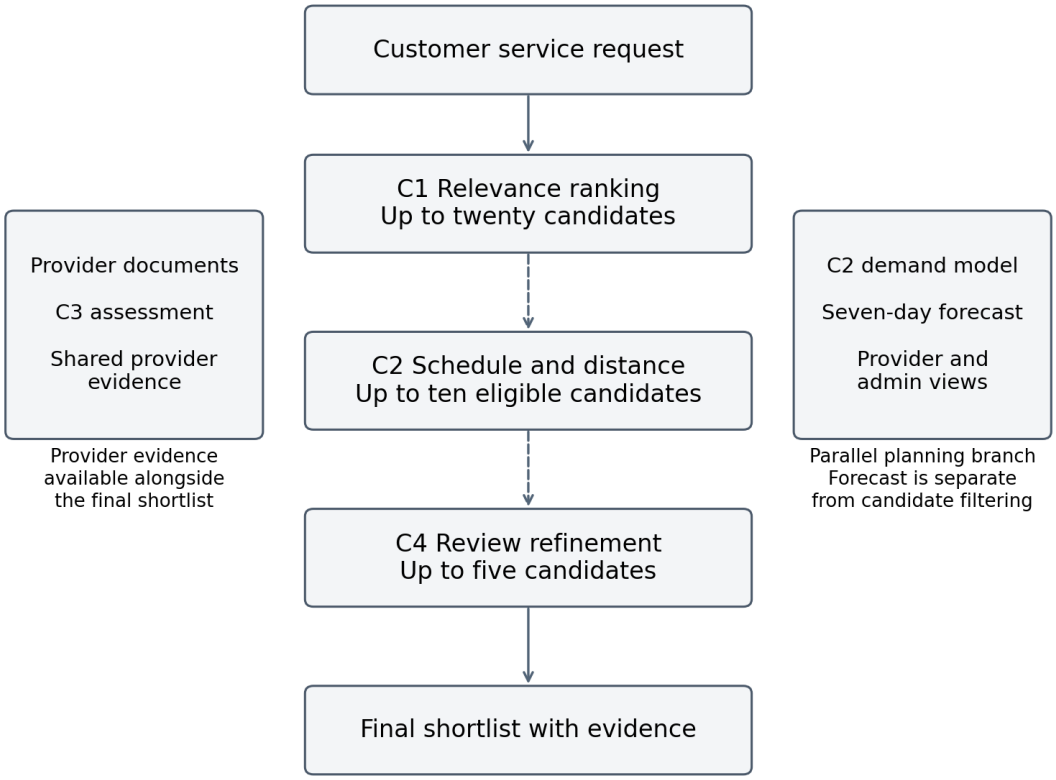


Figure 3.2 Intended candidate sequence and supporting branches

When a customer submits a requirement, C1 ranks profiles and returns up to twenty candidates. C2 should receive those same identifiers with the requested date, time and location. It calculates schedule eligibility and distance, attaches weather context and returns up to ten. C4 should analyse reviews belonging only to those retained candidates and produce up to five final recommendations. This is the proposed orchestration of the reported cardinalities. A full trace must confirm that the filter actually received C1's output rather than using its fallback of all provider records.

Candidate counts are upper limits. If only three candidates remain eligible, the final response should contain no more than those three. The system should not fill a five-

provider display with excluded or unrelated records. The current component reports do not establish the exact implementation of this smaller-set policy or the behaviour when candidates have no reviews. Those cases belong in the integrated acceptance suite.

The forecast branch compiles category-level daily and weekly counts, stores the target period and exposes the information through provider and administrator views. It does not alter the current request filter. After a real service is completed, a future operational feedback cycle could supply new interaction and review records. The common report does not claim that a complete booking-to-feedback cycle was measured in the supplied experiments.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The integration strategy is to preserve independent analytical responsibilities while exchanging results through shared application records and explicit candidate handoffs. C3 writes provider evidence, C2 writes demand summaries and request-specific filtering output, and C1 and C4 provide successive ranking outputs. This arrangement is compatible with the Firebase-based application interfaces described in the theses [1]–[4]. It reduces the need for one component to reproduce another component's internal model or feature calculations.

Integration must begin with common identities and categories. A provider should retain the same application identifier in onboarding documents, recommendations, filter results and reviews. The model-specific identifiers or row positions used during experimentation should not replace that identifier. A request should similarly retain one application identity from submission through final display. These shared keys establish the associations needed to interpret successive outputs.

The practical integration sequence should validate the provider and review records first, then the individual service interfaces, then adjacent candidate handoffs, and finally the complete customer journey. This is the recommended completion sequence based on the documented boundaries. The supplied materials do not include dated sprint records establishing that the team followed this exact sequence. Recorded component demonstrations are reported in Chapter 6 without converting this proposed sequence into a historical project-management claim.

The main integration risks are silent category defaults, stale results, missing scores, ambiguous credibility fields and incomplete worker state. Each risk has a concrete owner. The producing component owns the meaning and validity of its fields; the consuming component owns validation of received data; the shared application owns orchestration, display and access control. Appendix A specifies the responsibilities that should be confirmed in the next integrated release.

4.2 Interfaces data contracts and shared models

Table 4.1 summarises the documented interfaces and the additional common contract required at each boundary. Generic paths and field names describe the application structure without exposing live endpoints, personal identifiers or document-access links. Proposed metadata are explicitly distinguished from fields reported in the current implementation.

Table 4.1 Component interfaces and common contract requirements

Boundary	Documented input and output	Required integration control
C1 to candidate processing	Natural-language request provider data and Top-20 ranking.	Preserve request identity provider IDs hybrid score and original rank.
C2 filter	Request date time location candidate IDs and output_results.	Require supplied candidates validate intervals and prevent fallback expansion.
C3 provider record	Document groups and extractedFeatures with nested credibility.	Preserve original and model categories and separate processing from verification.
C4 review refinement	Up to ten candidates associated reviews and Top-5 reranking.	Preserve candidate membership review associations and aggregation configuration.
C2 demand views	daily_demand/current with daily values and category summaries.	Validate target week category mapping freshness and scenario assumptions.
Shared application	Provider customer review and request records.	Assign schema ownership versioning access rules and audit responsibility.

C2's current request representation includes user_id, service_date, service_time with start_time and end_time, location_type and isNewRequest. Its output_results contains provider_ids, evaluated_providers, weather_risk, recommendation, weather_summary and evaluated_at [2]. The filter can take candidates from supplied identifiers or existing result nodes, but can also fall back to all provider keys. In the integrated ranking flow, that fallback should be disabled or represented as a separate operating mode because it can introduce providers that C1 did not select.

C3 groups uploaded evidence into businessRegistration, certification, experienceProof, identityDocument and portfolioWork. Its output nests credibilityScore, credibilityLevel and lastEvaluatedAt within the extractedFeatures credibility object [3]. The existing payload should be preserved accurately when documenting current behaviour. A stronger contract should additionally distinguish directly extracted fields from heuristics and record the normalized model category, model version and evidence outcome.

C1's hybrid score and C4's trust-aware score have no documented common numerical calibration. The common report therefore does not add them together or invent a weighted final formula. An implementation that combines them must define its scales, missing-value policy and coefficients, then evaluate the resulting ranking. Otherwise, the application should present stage-specific scores with their meanings rather than imply that differently generated values are interchangeable.

Category harmonisation is a shared requirement. The demand vocabulary uses A/C Services and CCTV Installation, while the review vocabulary includes A/C Technicians and CCTV Technicians. The document model supports AC Repair and several appliance-repair categories, but not every category represented by the fourteen-category components [2]–[4]. Appendix A gives illustrative mappings and specifies that unsupported model coverage must remain explicit.

4.3 Shared nonfunctional requirements

Performance must be measured for the complete user-visible request, including profile retrieval, embedding computation, filtering, weather acquisition, review retrieval, neural inference and response display. The demand thesis reports model-search and batch-prediction times, but those values exclude most of this path [2]. They cannot serve as a group response-time guarantee. A controlled benchmark should record hardware, dataset size, concurrent requests, warm and cold model loading, and stage-level timings before a service target is adopted.

Reliability requires explicit worker states and repeatable outputs. C2 writes filtering output and request status separately, while C3 changes processing and verification flags after scoring [2], [3]. Firebase distinguishes partial updates and transactions in

its administrative interface [14]. The platform can use those mechanisms as part of a stronger state design, but the common report does not claim that atomic job claiming or idempotent retry handling is already implemented. A result must retain its request association and failure state even when a later stage cannot complete.

Security and privacy concern both the frontend and the analytical workers. Customers should not be able to change another provider's evidence or overwrite administrative assessment fields. Raw identity and qualification documents should be restricted to the authorised processing and review roles. Public report examples should use generic identifiers and omit credentials and access-bearing URLs. Current configuration and database rules require a dedicated review before deployment; model accuracy provides no evidence of access-control correctness.

Usability and accessibility require understandable stage labels, readable evidence summaries and a way to identify unavailable information. The interface should distinguish predicted demand from observed bookings, schedule eligibility from appointment confirmation, and processing completion from authenticity verification. Keyboard navigation, clear focus indicators, legible contrast and text alternatives for visual information are appropriate acceptance requirements. No measured accessibility or full-interface usability result is provided in the four theses.

Maintainability requires versioned models, schemas and preprocessing. C2 saves an encoder and ordered feature list, while C3 reconstructs categorical encoders in inference [2], [3]. C4 must retain its training tokenizer and sequence configuration, and C1 should retain the embedding model identifier and provider representation. The group release should package these artefacts consistently and identify which version produced each stored result.

4.4 Integrated workflow and exception handling

In normal intended operation, the application validates the request, obtains C1 candidates, passes their identifiers to C2, and supplies the retained set to C4. The final response must belong to the initiating request and contain only members of the accepted candidate lineage. Associated document evidence can be displayed

separately from the ranking stages. The forecast view can be loaded independently with its own target period and generation metadata.

The current filter exposes several failure paths that matter to this flow. A request for Monday 17:00–10:00 can be accepted by the interval-containment rule even though its end precedes its start. Unparseable times can also be accepted when the provider's working day is open. Missing provider coordinates receive no distance and may still enter a short retained list. When the requested date is absent from weather data, the helper can substitute the first forecast day without preserving that fallback indication in the returned summary [2]. These are known component behaviours, not successful integrated exception handling.

The strengthened workflow should reject reversed or malformed intervals before candidate evaluation. It should return a clear location or forecast-unavailable state when relevant data are absent, and distinguish records with unresolved distance from ranked candidates. Weather acquisition failure should not manufacture a low-risk advisory. Since the weather rules are advisory, the application can continue with explicitly unavailable context if the selected operating policy permits it, but that policy must be visible and consistent.

Document failure requires a separate treatment. A failed download, unreadable certificate or unsupported category should not lead directly to verified flags. C3's current normalizer maps A/C and unknown values to Electrician, and its rules can infer verification indicators from document presence [3]. A revised contract should return reviewRequired or extractionFailed while preserving the original evidence and category. Administrative verification should remain a distinct decision.

C4 needs an explicit missing-review policy. The supplied thesis describes review aggregation but does not establish the exact response when a candidate has no usable reviews or when review inference fails [4]. A sensible design would preserve the candidate with an insufficient-review-evidence state and an agreed fallback rank. That is a proposed design choice, not a documented feature. Replenishing the final list with providers excluded by the schedule stage would require a new retrieval cycle rather than silently bypassing eligibility.

4.5 Team contribution and dependency matrix

Table 4.2 maps the component owners identified in the supplied theses to their research and development responsibilities. The demand/filtering member is left as a specific metadata field because the submitted document contains no name or student ID. The functional content for that component is complete. Component responsibility should be confirmed against the team's contribution declaration before members sign the final report.

Table 4.2 Team contribution and dependency matrix

Member and component	Research and development contribution	Main dependencies
Hewapathirana N.T. IT22371836 C1	Lexical semantic and credibility-aware recommendation fusion and ranking evaluation.	Provider descriptions interaction history common categories and candidate consumers.
Masakorala A H IT22190048 C2	Synthetic demand experiment scenario inference schedule distance and weather processing.	Provider schedules customer location C1 candidate identifiers forecast views.
EMNS Roopasinghe IT22117182 C3	Synthetic policy-score comparison PDF and OCR processing evidence features and provider update.	Document storage provider category vocabulary extraction dependencies and administrative review.
Tharushika W.A.M. IT22914200 C4	Multi-task sentiment classification review anomaly assessment and provider reranking.	Filtered candidates provider-associated reviews tokenizer and aggregation policy.
Shared group work	Common application integration contracts evidence reconciliation and complete-system validation.	Agreed ownership release history test records and supervisor review.

The matrix assigns component work from the individual theses. Ownership of shared frontend, authentication, registration and database modules requires team confirmation. Planner, Teams, repository and diary records were not supplied. Appendix B provides the corresponding contribution declaration.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The project uses a Python analytical layer alongside a web application and shared data services. pandas and NumPy support structured data handling. scikit-learn supports text vectorisation, splitting, model selection, regression metrics and anomaly analysis. Sentence Transformers supplies the multilingual semantic representation for C1. XGBoost supports the selected regressors in C2 and C3. TensorFlow and Keras support C4's multi-task neural model [1]–[4]. The stack reflects different task requirements rather than one uniform model architecture.

C1's representation uses TF-IDF vectors and sentence embeddings. C2's representation combines generated tabular features with a fitted target encoder. C3 uses encoded service and qualification categories with evidence counts and numerical heuristics. C4 uses tokenized review sequences and a separate credibility representation. These preprocessing artefacts belong to their respective models. A shared deployment should not substitute one component's encoder, category index or tokenization procedure for another's.

Firebase is the common application data service described in the implementation accounts. Its administrative interface is used by Python workers, while the demand thesis describes React and TypeScript consumers with Firebase client functions and Vite tooling [2]. The application includes demand and filtering-history views. The common architecture therefore has a client boundary and a privileged worker boundary, even though the supplied theses do not document one locked deployment environment covering all services.

C3 also depends on Poppler, Tesseract and optional scanned-PDF rasterisation support. C2 uses HTTP weather acquisition and external requests. These dependencies need separate readiness checks. A model may be loadable while OCR is unavailable or the weather service cannot be reached. The release should report those states explicitly rather than equate general application availability with availability of every evidence-processing capability.

5.2 Development and integration process

The individual reports describe model preparation, experimentation and modular implementation. C1 compares separate matching methods before hybrid fusion. C2 preserves model, encoder and feature order for forecasting. C3 separates local document inference from a database integration worker. C4 develops sentiment and review-credibility processes independently before provider aggregation [1]–[4]. These activities provide the component-level development foundation for the common solution.

The common report synthesises the interfaces after examining those implementations. It reconciles the Top-20, up-to-10 and Top-5 stages, distinguishes three credibility concepts, and identifies category and database inconsistencies. This documentation work produces a coherent integration specification; it does not establish that every discrepancy has already been corrected in code. The next implementation cycle should apply the agreed contracts to the actual services and retain evidence for each correction.

Version control and code review should connect component changes to a common release. Each change that affects a category map, request schema, score interpretation or state transition should identify its producer and affected consumers. Review should include contract examples and regression cases, especially where a silent fallback could produce a plausible but incorrect output. These are proposed shared development controls. Although the C1 thesis lists Git/GitHub, no commit history or continuous-integration log is attached [1].

The release procedure should package preprocessing and model artefacts together, verify local dependencies, exercise protected data access and run adjacent-stage tests before the complete journey. A smaller fixture with agreed provider IDs and manually known expected outputs is appropriate for deterministic integration checks. Model-performance experiments should remain separate so that a pipeline test is not mistaken for a fresh evaluation of ranking quality or forecasting generalisation.

5.3 Deployment and operational configuration

The available evidence supports research notebooks, saved model artefacts, frontend implementation descriptions and selected historical database operations. It does not establish a production deployment with measured uptime, automated rollback, backups or a complete recovery test. C3's continuous polling activation is commented out in its source account, while C2's recorded filtering listener was idle and manually terminated [2], [3]. These details constrain operational claims despite the presence of worker functions.

An integrated deployment should separate the public client from privileged analytical services. Administrative credentials should be supplied through controlled configuration and should not be bundled with client code or distributed component archives. Document downloads require permitted sources, file-type and size validation, timeouts and temporary-file cleanup. The model and tokenizer artefacts should come from the trusted release package. Exact hosting choices and service endpoints require the team's confirmed deployment manifest.

Demand generation should record the compile date, target week and input assumptions. The current forecast location named current is convenient for the web view, but a retained history is needed to distinguish releases and evaluate forecasts when later observations become available [2]. Similarly, reassessment of a provider should retain the old result, evidence change and model version rather than overwrite an unexplained score. Request results should retain both the stage outputs and the final response association.

Monitoring should identify model loading failure, extraction failure, weather timeout, invalid request, missing review data and database write failure as different events. Logs should preserve operational context without copying sensitive document contents or full personal records. Recovery tests should demonstrate that retrying a job does not duplicate effects, assign a result to another request or change verification status solely because processing resumed. No such complete recovery result is supplied; it is part of the deployment work required before a controlled pilot.

5.4 Evidence of team execution

The available evidence of execution is concentrated in the four individual theses. C1 reports functional tests of request handling, score generation, normalization, Top-20 ranking and Firebase retrieval. C2 reports model comparisons, forecast consistency, a historical demand write and an idle listener. C3 reports regression experiments, four document fixtures and one provider update. C4 reports train/validation/test partitions, sentiment and credibility metrics, and the candidate-reranking process [1]–[4]. These are concrete contributions to the group prototype.

Their evidence strength differs. Numerical model tables establish the reported experiments. Screenshots and demonstration outputs can establish selected processing steps. Descriptions of a workflow establish the intended design but do not necessarily show an observed result for every cross-component boundary. The common report preserves this distinction when presenting overall results. It does not convert the number of model experiments into a count of complete integrated system tests.

No MS Planner export, Teams meeting record, commit log or research diary was attached. The report therefore cannot provide dated claims about sprint completion, meeting attendance or continuous integration. Appendix B records the component allocation supported by the named reports and identifies shared responsibilities for confirmation. This gives reviewers a transparent team structure while leaving activity claims tied to records that the team can supply if required.

The common documentation itself contributes a reconciled architecture and evaluation framework. Its diagrams and interface descriptions are newly prepared syntheses of the supplied evidence. They should be reviewed against the release code before being used as as-built deployment diagrams. Appendix C records the source differences and the AI assistance used in report preparation, ensuring that the provenance of this common artefact remains clear.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The group evaluation must answer two different questions: whether each analytical component performs its defined task, and whether the components preserve correctness when combined. The individual studies answer parts of the first question. The common report assesses the second through the documented interfaces and available integration evidence. A complete four-component test has not been newly executed for this report, and no aggregate system-accuracy value is assigned.

For C1, the reported outcome is category-level ranking measured with Precision@K and normalized discounted cumulative gain (NDCG@K) at K values of 5, 10 and 20. Precision considers relevant items in the returned set, while NDCG also reflects their positions [15]. C2 measures regression error and derived demand bands on synthetic records; C3 measures agreement with a defined score policy; C4 measures aspect classification and review-credibility labels [1]–[4]. Their experimental units and targets differ, so each metric retains its own denominator and meaning.

The common acceptance criteria proposed here concern identity, lineage, validity and interpretation. A final provider must belong to the accepted upstream candidate set. Every stage output must be associated with the initiating request. An unsupported category, reversed interval, unavailable forecast or failed extraction must produce an explicit state. Models must receive the preprocessing representation used during training. These criteria can be assessed with deterministic fixtures independently of numerical model performance.

For subsequent performance validation, the group should compare final Top-5 rankings with the initial C1 ranking and a context-filtered baseline using the same requests and independent relevance judgements. It should record eligibility errors, final-list size, response time and failure recovery. Forecast evaluation should use later observed bookings and a rolling origin [16]. Provider evidence should be compared with manually labelled fields and a justified review rubric. Stakeholder tasks should evaluate whether users interpret outputs correctly. These are future evaluation procedures, not completed results.

6.2 Integration and end to end testing

Table 6.1 identifies what the submitted evidence supports and what the complete workflow still needs. Reported component tests are attributed to their source theses. A test requirement without an observed result is not labelled Pass. This approach allows the common report to be complete as an evidence account without implying that missing release tests have already run.

Table 6.1 Available testing evidence and complete workflow requirements

Check	Available evidence	Group interpretation
Initial ranking	C1 reports functional tests and Top-20 output.	Reported component capability.
Demand aggregation	C2 reports consistent 98-row and 14-summary CSV outputs.	Scenario consistency demonstrated.
Filter input validation	C2 helper checks expose reversed and unparseable time acceptance.	Known issue requires correction.
Provider evidence	C3 reports four local fixtures and one provider update.	Selected demonstrations only.
Review processing	C4 reports test metrics and provider aggregation workflow.	Analytical capability reported.
Candidate handoff	Top-20 Top-10 and Top-5 stages appear in different theses.	Full same-request trace not supplied.
Shared category handling	C3 A/C default and differing category vocabularies are documented.	Mapping correction and contract tests required.
Worker recovery	Separate writes and polling are described.	Concurrent and interrupted processing not established.
Access and usability	Security and user-study requirements are discussed.	Complete-system validation results not supplied.

The recommendation thesis reports ten functional scenarios marked Pass, including lexical and semantic score generation, normalization, missing historical information, ranking, Top-20 output and application-data retrieval [1]. Those results are retained

as reported C1 tests. They do not establish that the same request subsequently passed through C2 and C4, or that final display preserved all eligibility constraints.

The demand component's historical synchronization writes a forecast compiled on 16 August 2026 for 17–23 August and subsequently reads fourteen category entries. Its filtering listener reads 64 request nodes and finds zero new requests [2]. Consequently, the listener demonstrates connectivity and an idle polling pass, rather than a completed newly submitted filter request. The document component separately records a successful update of one provider [3]. These operations belong to different data paths and should not be described as one end-to-end run.

C4 describes integration from candidate receipt through review processing and reranking, but no common request trace links that output to the precise C1 candidate list and C2 result used in an integrated release [4]. The next test should retain anonymised request input, stage outputs, timestamps, candidate membership and the final displayed response. Failure cases should repeat the trace with invalid intervals, missing reviews, weather unavailability and interrupted writes. A test harness should verify the exact output membership rather than only the absence of an exception.

6.3 Overall results

The overall result is a four-component research prototype with reported analytical performance and selected application demonstrations. The components collectively provide relevant retrieval, declared-context filtering, provider-evidence processing, review analysis and category-level planning. Their combined design is coherent, but the evidence does not yet establish measured improvement of the final customer decision over a complete baseline system.

C1 reports datasets containing 10,000 providers, 20,000 requests and 100,000 interactions, with 500 sampled requests used for the ranking comparison [1]. The thesis describes these as prepared or controlled study materials in relevant passages; collection and sampling provenance would need a fuller artefact record before treating them as a representative operational sample. Table 6.2 reports the comparable K=20 values from the thesis's extended results. Figure 6.1 redraws those values without adding new experimental observations.

Table 6.2 Reported recommendation comparison at twenty candidates

Method	Precision at 20	NDCG at 20
TF-IDF	0.9231	0.9297
Multilingual semantic matching	0.9387	0.9410
Credibility collaborative score	0.0672	0.0654
Hybrid	0.9442	0.9458

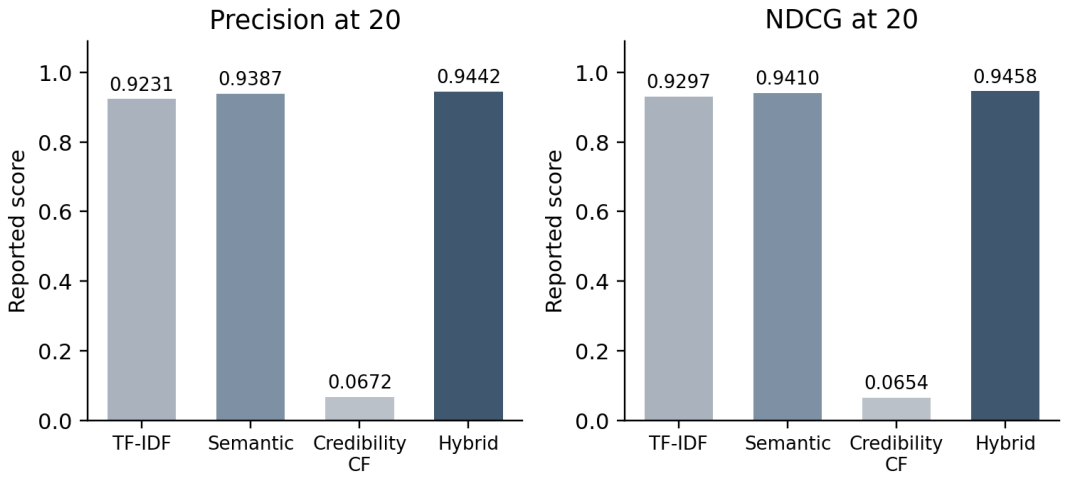


Figure 6.1 Reported category relevance comparison for Component 1

The hybrid result exceeds the strongest standalone semantic result by 0.0055 in Precision@20, or 0.55 percentage points. It also reports Precision@5 of 0.9490 and Precision@10 of 0.9483, with corresponding NDCG values of 0.9496 and 0.9490 [1]. These are incremental improvements under the category-based definition. The low standalone collaborative value reflects that rating, activity and history alone do not identify the request's service category. The result does not establish that the document-based score improves ranking, because that score is not the documented input to C1's collaborative calculation.

Table 6.3 places the two regression experiments alongside one another while preserving their units. The demand error is measured in generated order counts; the document error is measured in policy-score points. These RMSE values cannot be used to rank the two components against each other. Both experiments select XGBoost within their own reported comparison [2], [3].

Table 6.3 Reported regression outcomes with separate target meanings

Target and unit	Model	RMSE	R squared
Demand count generated orders	Random Forest	3.454984	0.880715
Demand count generated orders	XGBoost	3.423339	0.882890
Document policy score points	Random Forest	1.569	0.989
Document policy score points	XGBoost	1.141	0.994

For demand, XGBoost's MAE is 2.926993 and mean absolute percentage error (MAPE) is 26.049276%, while Random Forest's MAE is 2.967926 and MAPE is 25.201550%. The selected model has lower absolute error but higher percentage error. Its RMSE advantage is approximately 0.92% relative to Random Forest. A category training-mean baseline achieves RMSE 3.659218 and R squared 0.866196, showing that much of the synthetic task can be explained by category structure [2]. The demand-band result is 476 correctly assigned bands among 600 holdout rows, or 79.33%.

For provider scoring, the known formula reproduces all 2,000 generated targets with zero RMSE. This is a direct policy baseline rather than evidence of perfect real-world reliability prediction. Identity is constant in training, and 33.80% of generated scores reach the upper cap [3]. The reported XGBoost fit demonstrates approximation of the defined target. Four document fixtures and the separate 84.60 integration result demonstrate selected processing paths without establishing OCR accuracy or document authenticity.

C4's sentiment results are shown in Table 6.4 and Figure 6.2. The average accuracy of 88.94% is the mean of four aspect results from the same review study. It is not the accuracy of the complete platform, and it does not represent four independent user studies. Lower accuracy and F1 for Communication and Professionalism indicate where provider explanations should avoid treating all aspect predictions as equally reliable [4].

Table 6.4 Reported aspect sentiment test results

Aspect	Accuracy percent	F1 percent
Quality	96.10	96.11
Punctuality	87.73	87.61
Communication	86.35	86.16
Professionalism	85.58	85.28

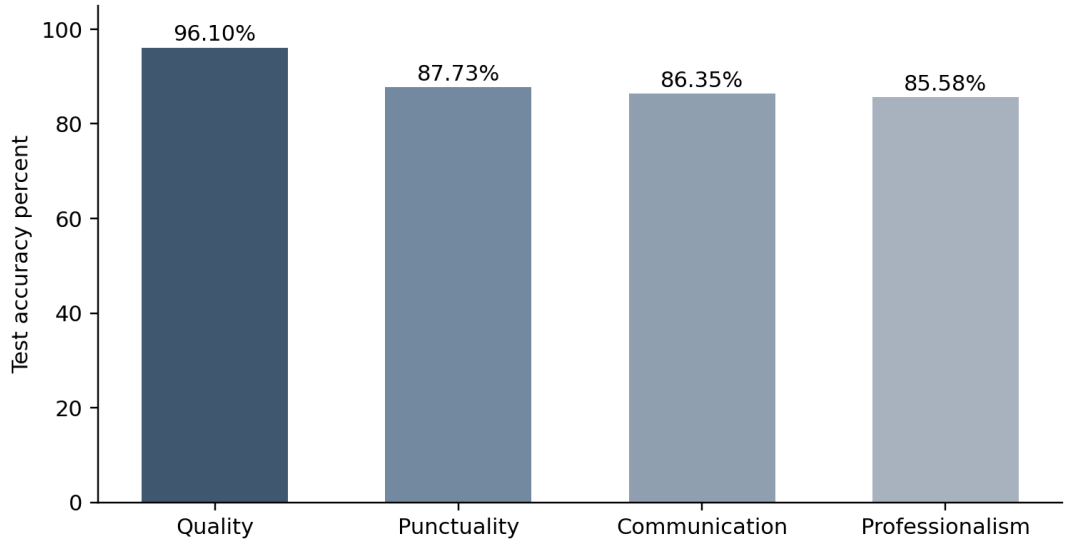


Figure 6.2 Reported accuracy for the four service review aspects

The review-credibility test has 3,387 records labelled Real and 386 labelled Fake. Table 6.5 retains the source labels for reporting while interpreting the operational output as anomaly-related evidence. Overall accuracy is reported as 96%, but the minority-class F1 is 0.79 and macro F1 is 0.88 [4]. From the reported supports, always predicting the majority label would give 3,387 divided by 3,773, or approximately 89.77% accuracy. That value is an arithmetic reference computed here, not a newly trained baseline experiment. It illustrates why the class-specific measures matter.

Table 6.5 Reported review credibility classification results

Source label	Precision	Recall	F1	Support

Source label	Precision	Recall	F1	Support
Real	0.98	0.97	0.98	3387
Fake	0.77	0.80	0.79	386
Macro average	0.87	0.89	0.88	3773
Weighted average	0.96	0.96	0.96	3773

At group level, these results support the availability of distinct analytical capabilities. They also identify bottlenecks that cannot be corrected by averaging accuracies: malformed schedule inputs, incomplete document interpretation, ambiguous score provenance and missing complete-workflow evidence. The integrated value must be established by comparing final recommendations and user understanding under a common request-based evaluation.

6.4 Stakeholder and user validation

The supplied documents focus primarily on technical and experimental evaluation. C1 states that a formal large-scale user study is outside its scope. C2 and C3 report no completed stakeholder dataset. C4 identifies broader user validation as future work [1]–[4]. The common report therefore cannot provide participant counts, satisfaction percentages or acceptance-test outcomes for the complete product.

A controlled stakeholder evaluation should involve customers, providers and administrators performing different tasks. Customers could compare final recommendations, explain why a provider is shown and distinguish eligibility from booking confirmation. Providers could identify inaccurate evidence and interpret a forecast's target period. Administrators could respond to failed extraction, category conflicts and unavailable weather. The evaluation should record task completion, interpretation errors, time, feedback and the changes made after review.

Evaluation should also include users and providers who may be disadvantaged by the current assumptions. New providers may have little interaction history, applicants may lack formal certificates, and reviews may use vocabulary or languages underrepresented in training. These conditions should be represented deliberately

rather than excluded by selecting only complete profiles. Recruitment, consent and institutional review requirements should be settled before collecting new data. This report presents the validation design without claiming that those activities occurred.

6.5 Achievement of common objectives

Table 6.6 evaluates the six common objectives against available evidence. Achieved capability is qualified by the setting in which it was demonstrated. Partial achievement identifies a concrete missing link, rather than implying that an entire component failed. The assessment concerns the submitted project evidence and the common synthesis, not a newly executed release acceptance test.

Table 6.6 Achievement of the common objectives

Objective	Assessment	Evidence and remaining boundary
G1 Request relevance	Achieved in reported component evaluation.	Hybrid ranking exceeds individual methods under category relevance.
G2 Context and planning	Partly achieved.	Forecast scenarios and filtering rules exist; robust input handling and live forecast validity remain open.
G3 Provider evidence	Achieved at prototype level.	Features scores and database update demonstrated; authenticity and extraction accuracy unvalidated.
G4 Review refinement	Partly achieved at group level.	Aspect and credibility models evaluated; final ranking benefit lacks a common-system comparison.
G5 Integration consistency	Partly achieved.	Common contracts and intended sequence defined; same-request trace and mappings still require verification.
G6 Responsible evaluation	Achieved in this report.	Results units evidence classes limitations and proposed validation are distinguished.

The strongest integrated achievement is the definition of a modular decision workflow with explicit evidence boundaries. The team has reported concrete implementations and numerical evaluations, and the common report makes their relationships inspectable. The remaining work centres on verified orchestration and

independent outcome evaluation. Successful component demonstrations provide a basis for that work, but cannot replace it.

6.6 Integrated limitations and threats

The first limitation is the difference between component targets and the intended user outcome. Category agreement tests retrieval relevance, synthetic counts test the demand generator, and synthetic credibility labels test a policy. Sentiment accuracy tests review interpretation. None alone establishes that a customer selected the most suitable professional or received better service. A final workflow study needs its own outcome definition and independent evaluation data.

Validation design also affects interpretation. C2 fits target encoding on the complete training partition before its inner search, creating a cross-validation leakage concern within that search, while holdout labels are not used to fit the encoder [2]. Data-dependent preprocessing should be fitted within the appropriate training fold [17], [18]. Its random split is not a rolling future-forecast test [16]. C3 uses the holdout to select a model family and reuses it in a training-fraction diagnostic [3]. A further untouched assessment set would strengthen both comparisons.

The review dataset has different performance across aspects and imbalanced credibility labels. The supplied report does not establish that provider identities, time periods or duplicated text were fully isolated across partitions, so independent-provider or later-period generalisation remains to be evaluated [4]. C1's category-based labels do not fully test semantic specialisation or provider reliability, and its controlled data provenance requires fuller documentation [1].

Integration threats include C3's silent category defaults, the absence of identical category coverage across models, missing candidate lineage, and C2's time and weather fallbacks. A precise numerical output can be attached to the wrong semantic input or request without raising an exception. The three credibility meanings can also be confused if the frontend reuses one generic label. These threats concern the correctness of the complete system even when individual inference functions operate normally.

Operational limitations include incomplete environment locking, reconstructed preprocessing, separate status writes and unmeasured load or recovery behaviour. User validation, security testing, accessibility and measured commercial feasibility are not established. These limitations support a controlled prototype conclusion and define the priorities in Chapter 8. They should be resolved before claiming that the integrated platform is ready for dependable public operation.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The proposed product value is a clearer provider-selection process within one application. Customers could move from a service description to a relevant and context-aware shortlist, then inspect supporting provider and review information. Providers could present qualifications and experience in a structured form and view category-level planning scenarios. Administrators could inspect processing and uncertainty rather than manage disconnected documents and model outputs.

This value arises from the relationships among the components. C1 reduces the initial search space, C2 adds declared operating context, C3 structures onboarding evidence, and C4 provides finer review interpretation. A combined user interface can make those signals easier to understand than four independent outputs. The current evidence supports the technical feasibility of the component methods, while reduced selection time, improved trust and increased booking success remain hypotheses requiring stakeholder and workflow evaluation [1]–[4].

The initial beneficiaries should be defined narrowly for a controlled pilot. A small set of supported service categories, complete provider records and reviewed evidence would allow the team to evaluate errors and correction needs before expanding. The product should allow informed customer choice and provider appeal rather than present the final ranking as a guaranteed judgement of workmanship.

7.2 Market alternatives and differentiation

Relevant substitutes include manual referrals, searchable directories, category-and-rating filters, manual document review and platforms that present aggregate reviews. These alternatives solve different portions of the selection problem. The proposed differentiation is the connection of natural-language matching, declared schedule and location checks, document evidence and aspect-level review analysis in one workflow. This is a defensible architectural distinction from a simple listing, but no verified current market comparison establishes that the project is unique or superior to every existing platform.

The project also has simpler technical alternatives that should remain baselines. TF-IDF or semantic retrieval alone can generate candidates. Validated deterministic rules can perform eligibility checks. C3's known policy can be calculated directly, and aggregate ratings can be compared with C4's aspect-based refinement. The recorded C1 improvement over the semantic baseline is incremental, and C3's learned model cannot outperform the exact formula on the generated policy labels [1], [3].

Commercial evaluation should compare the complete product with the actual workflow used by the target participants. Relevant criteria include task time, explanation quality, eligibility errors, correction burden, final recommendation usefulness and operating cost. Such evaluation would identify whether integration creates practical benefit beyond extra model complexity. No pricing, market share, named competitor performance or acquisition rate is asserted in this report.

7.3 Business model and feasibility

A potential sustainability model is a platform service funded through marketplace operation or an institutional pilot. Future revenue could involve provider subscriptions or a clearly disclosed service fee, but those arrangements are proposals rather than agreed project terms. Payment should not influence a credibility score or an undisclosed ranking advantage. Any promoted placement should be labelled and evaluated separately from the research ranking.

The principal resources are application hosting, database operations, document storage, OCR processing, semantic and neural inference, model maintenance and administrative review. Weather acquisition and review processing introduce external-service or computation dependencies. Costs should be measured per processed onboarding case and per completed request rather than inferred from one batch-prediction time. No cost-per-request, revenue or break-even estimate is available in the submitted theses.

Technical feasibility is supported at component level through model results and selected demonstrations. Operational feasibility is less established because safe worker recovery, complete orchestration and realistic load testing remain open.

Human review is particularly important for ambiguous documents and contested evidence. A sustainable pilot should measure the fraction of cases requiring intervention, time to resolve them, and whether automation reduces total administrative effort while retaining acceptable accuracy.

Adoption channels could include the participating providers and customers of a controlled university-supported deployment. Expansion should follow demonstrated performance in the supported categories and languages. An initial bounded pilot would provide genuine interaction, booking and review data for later research, subject to appropriate data-handling arrangements. It should not treat synthetic demand scenarios as a basis for revenue forecasts or promised provider earnings.

7.4 Adoption intellectual property and compliance

Adoption depends on accurate information, clear explanations and correction routes. Providers may reasonably question a score derived from a misread certificate, unsupported category or overlapping experience letters. Customers may misunderstand a high sentiment result as a service guarantee. The interface should show the relevant evidence and uncertainty, and administrators should be able to correct extraction and review decisions without altering unrelated profile data.

The team should document ownership of the original code, datasets, trained artefacts, report figures and shared application work. Third-party software and pretrained models have separate licences that must be respected. A release inventory should distinguish project-developed logic from established libraries and model weights. The open availability of software does not make uploaded personal documents freely distributable or permit an unsupported claim that the team created the underlying algorithms.

Personal-data handling requires a deployment-specific review. Sri Lanka's Data Protection Authority describes its mandate as regulating personal-data processing and safeguarding data subjects' privacy [19]. The platform should use that official framework when reviewing identity documents, account information, review records, retention and access arrangements. This report does not certify legal compliance or

state an unverified legal commencement date. A technical demonstration alone cannot establish that every institutional and regulatory requirement has been met.

Responsible use also requires separating anomaly evidence from accusations. A review identified as unusual may still describe a legitimate experience. Document presence may indicate submitted evidence without establishing authenticity. The final ranking should preserve these meanings, provide review and correction opportunities, and monitor adverse effects on new or less-documented providers. The AI assistance used for this report is disclosed in Appendix C, with final responsibility retained by the team.

7.5 Sustainable Development Goal contribution

The plausible SDG 8 pathway is improved access to productive service opportunities through clearer matching and provider representation [5]. A pilot could measure completed matches, provider participation, repeated service use and the distribution of opportunities. It should also ask whether the platform creates barriers for people with sparse digital histories or limited documentation. Increased platform activity alone would not establish decent work or improved working conditions.

The SDG 9 pathway concerns a modular digital service capability that can be maintained and evaluated [6]. Relevant measures include supported service coverage, successful workflow completion, processing reliability and access to understandable information. These indicators would assess the actual digital service rather than use the presence of machine learning as evidence of social impact.

Environmental and operational sustainability favour processing only changed evidence, retaining documents for a justified period and avoiding unnecessary repeated inference. Caching provider representations and reusing valid review analysis could reduce resource consumption, provided version and freshness controls remain explicit. No energy or carbon measurement is supplied, so no quantified environmental benefit is claimed. Sustainable outcomes must be evaluated alongside inclusion, privacy and correction costs.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The project addresses maintenance-provider selection by connecting four forms of information: request relevance, declared operational context, onboarding evidence and customer-review intelligence. The reported components provide initial Top-20 retrieval, up-to-ten schedule and distance candidates, structured document scores, and a review-based Top-5 refinement design. Category-level forecasting supports provider and administrator planning alongside the request flow [1]–[4].

The individual studies report useful numerical results: hybrid Precision@20 of 0.9442, demand-regression R squared of 0.882890, document-policy R squared of 0.994, and average aspect sentiment accuracy of 88.94%. These values describe different experimental tasks. They support component capability while leaving final recommendation benefit, observed future demand, authenticated provider evidence and complete-user acceptance to further evaluation.

The overall outcome is a coherent research prototype and an inspectable integration design. The common problem is addressed technically by connecting complementary decision stages. Stronger evidence that this connection improves real customer choices requires a versioned same-request trace, corrected category and state handling, and independent stakeholder and outcome validation. The report's conclusion is therefore proportional to the measured component work and the remaining integrated evidence.

8.2 Principal integrated contribution

The principal integrated contribution is a modular provider-selection framework that gives each signal a defined role. Lexical and semantic relevance generate candidates; operational rules refine declared suitability; documents support onboarding evidence; review models interpret past customer feedback. Shared identities and contracts make these outputs available within one application without collapsing them into an unexplained universal score.

The common synthesis also identifies where the components require stronger coordination. The Top-20, up-to-10 and Top-5 interfaces need preserved membership and score lineage. Document, interaction and review credibility must retain separate meanings. Demand forecasts need target periods and scenario provenance. These are practical findings that arise from considering the four components together and would be less visible in a collection of independent model summaries.

8.3 Future directions

The first priority is integration correctness. The team should agree the category catalogue, correct silent category defaults, validate request intervals and locations, preserve candidate lineage, and separate document processing from verification. The same release should define behaviour for fewer than five candidates, missing reviews and unavailable weather. A complete fixture trace should demonstrate these behaviours across the actual services and final interface.

The second priority is stronger research validation. Demand should be evaluated against later observed bookings with time-respecting preprocessing. Provider scoring needs independent evidence labels and a representative extraction benchmark. Recommendation and reranking should be compared on the same requests using richer relevance and service-outcome criteria. Review evaluation should examine later periods, unseen providers and language variation. Current component baselines should remain visible in those experiments.

The third priority is a controlled operational and stakeholder pilot. The release should include packaged models and preprocessing, safe credentials, access rules, retry handling, monitoring and recovery tests. Customer, provider and administrator tasks should assess correct interpretation as well as perceived usefulness. Expansion and commercial commitments should follow measured reliability, inclusion and cost rather than extrapolation from synthetic accuracy.

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APPENDICES

Appendix A Integrated interface specification

A.1 Shared identifiers and candidate lineage

The common contract proposed for the integrated release should use one request_id and the application provider_id throughout processing. C1 should supply original rank, hybrid score and its component scores with every candidate. C2 should preserve those values while adding schedule eligibility, distance and weather context. C4 should preserve membership and associate review evidence and its final ranking with the same provider. These additional lineage fields are proposed common metadata, not a claim that the existing filtering payload already contains them.

The required membership checks are that every C2 candidate belongs to the C1 set and every C4 result belongs to the C2 set. Duplicate provider identifiers should be rejected or resolved consistently. Candidate counts should be at most twenty, ten and five at the respective stages, without padding from unrelated records. A stage should record empty and insufficient-evidence outcomes explicitly.

A.2 Category vocabulary and model coverage

Table A.1 illustrates why a shared display catalogue needs component-specific mappings. It preserves the source terms rather than implying that all model vocabularies are identical. The table is a proposed mapping specification for confirmation against the release code. Appliance categories supported by C3 must not be treated as interchangeable with broad categories in another model without an agreed policy [2]–[4].

Table A.1 Illustrative category translations and coverage differences

Common service	Demand or review vocabulary	Document model vocabulary
Electrical services	Electricians	Electrician
Plumbing services	Plumbers	Plumber
Carpentry	Carpenters	Carpenter
Air conditioning	A/C Services or A/C Technicians	AC Repair

Common service	Demand or review vocabulary	Document model vocabulary
CCTV services	CCTV Installation or CCTV Technicians	No explicit CCTV category in the ten-category model.
Cleaning services	Cleaners	No explicit cleaning category in the ten-category model.
Appliance repair	Broad Technicians may require a more specific mapping.	Refrigerator Repair Washing Machine Repair TV and Electronics Repair.

The original label, canonical application label and model label should all remain inspectable when a translation occurs. Unsupported categories should return a coverage or review state. The A/C-to-Electrician fallback reported by C3 is a known implementation issue and is not the intended mapping in Table A.1.

A.3 Evidence scores forecasts and states

C3's model input has ten ordered fields: encoded service category, identity indicator, certificate count, encoded highest qualification, issuer-reputation heuristic, registration indicator, experience years, reference count, portfolio heuristic and portfolio count. The output score is bounded to 0–100, with thresholds at 60 and 85. The raw document outcome and authenticity decision should remain separate from those policy bands [3].

C2's forecast contract should preserve seven dates, fourteen categories, daily counts, total_weekly, avg_daily and demand_level, together with compile date and target period. Current generated counts are scenario predictions. Daily bands and a band derived from the weekly rounded average have different aggregation meanings. A client should not represent the payload as observed bookings or use cached data without indicating their source and period [2].

C4's contract should retain review_id, provider_id, aspect predictions, credibility evidence and the aggregation configuration. Its exact final formula and missing-review policy require confirmation before deterministic reranking tests [4]. Proposed worker states are pending, processing, completed, retryableFailure and reviewRequired. Verification should be recorded independently.

Appendix B Team contribution declaration

The component allocation in Table 4.2 is based on the submitted individual theses. Hewapathirana N.T. is associated with the hybrid recommendation engine, EMNS Roopasinghe with provider document scoring, and Tharushika W.A.M. with review sentiment and credibility. The demand/filtering contribution is described fully, but its owner's name and student ID must be added because that source still contains placeholders.

Each member should confirm the research tasks, implementation work and integration contribution attributed to them. Shared frontend, authentication, registration, database setup and release testing should be allocated by agreement and supported by available project records. This declaration is not a certification that those records were inspected during report preparation. The signature fields below are intentionally unsigned.

Table B.1 Team confirmation of component and shared contributions

Member	Student ID	Signature and date
Hewapathirana N.T.	IT22371836	_____
Masakorala A H	IT22190048	_____
EMNS Roopasinghe	IT22117182	_____
Tharushika W.A.M.	IT22914200	_____

Appendix C Source reconciliation and AI assistance

The project title, group R26-IT-072 and academic context follow the named recommendation and review theses. The demand/filtering member's name and ID remain unresolved. Supervisor spellings follow the review thesis: Dhammika De Silva and Samantha Rajapaksha. The document thesis uses Dammika and Rajapaksa; official spellings require team confirmation.

The database description uses Firebase as the common application boundary, while recording that several recommendation-methodology passages mention MongoDB. The candidate flow is an intended reconciliation of Top-20 retrieval, up-to-ten filtering and Top-5 review refinement. It is not proof of a measured full-system run. C1's activity-based credibility, C3's document policy score and C4's review anomaly evidence remain separate throughout the report.

C2's 3–9 August scenario CSV total of 1,926 and the separate 17–23 August demand-sync result are different dated artefacts. The reported Tile Workers value of 94 in the former CSV pair differs from a notebook value of 98. The common report uses the internally consistent CSV account without mixing it with the later period. C3's A/C normalization issue and processing-versus-verification issue are retained as limitations. C4's Real/Fake labels are used only when reporting the source test table, with anomaly interpretation maintained in the operational discussion [2]–[4].

ChatGPT with Codex assistance was used to review the four theses and template, reconcile sources, check primary references, draft the English text, create tables and synthesis diagrams, calculate arithmetic comparisons, and format and render the report. Report preparation involved no new experiment, model training, live database operation, interview or authentication study. Historical results remain attributed to the supplied theses.

The team should verify the technical account and contribution allocation, complete member metadata and signatures, and disclose further assistance required by the module policy. Proposed contracts and acceptance procedures should remain clearly distinguished from deployed code.